

# Laboratory Exercises – General Advises and Rules (v 0.2)

*based on the documentation from Peter Schulz*

Please read this document carefully and comply with all advises and rules mentioned, as this is a mandatory requirement for successfully pass the exercises.

## General information

- You are only allowed to take part in the lab sessions if you have taken part in the safety instructions.
- The laboratory instructions must be read and entirely understood

## Preparation

- All preparatory tasks must be completed and documented before the lab session starts.
- Also, it must be brought with you to the laboratory appointment.
- All team members must be able to answer questions regarding all preparation tasks
  - Insufficient preparation or answers will negatively affect the exercise's score for the team

## During the lab session

- Be on time.
  - The laboratory is a compulsory course.
  - You may miss important announcements and information.
  - It is also unfair to your team colleagues because you also must have the preparation tasks checked together with your colleagues. The start for the entire team would be delayed
- Electronic experimental setups and measuring devices can easily be damaged by incorrect operation. Hence, the following rules of conduct must be strictly adhered to:
  - Only connect the intended power supply to your experimental setup.
  - Do not touch the electrical connectors of the components used with your fingers.
  - Do not connect logical outputs to one another (short circuit), either by wire or through incorrect entries in constraints files.
- It is highly recommended to record all relevant steps and results, either on paper or by saving the appropriate files on your USB memory stick.
  - All files created and data modified on the lab PCs are deleted after the system is shut down and data restoration is only possible in exceptional cases

## Reports

- The entire team must prepare a report. It is a community effort.
- The key point when writing a lab report is that every reader must be able to understand what you did and what results you obtained so that they could repeat the same experiment.
- Pictures must be black (or color) on white background.
  - Avoid white (or color) on dark background
- The report must be submitted as pdf **within one week after the lab session** via TEAMS
  - Hard deadline/ no extensions
  - Missing the deadline will result in a 0 point score for the report
  - Even if no report is required (exercise 4), the title page has to be submitted
- Report structure and content **(5 pages + title and additional material at maximum)**
  - The front page **(1 page)**
    - Name of the course (Digital Circuits)
    - Name of the experiment
    - Group Number + Team Number in the form N.M (N: group, M: team/table)
    - Names and immatriculation numbers (IM-No.) of all participants
    - Date of lab performance
    - Score board of the respective exercise, aligned with the templates at the end of this document
  - Introduction **(~1/2 page)**
    - Summary of the exercise's objective in your own words
    - A copy of the lab task description is not acceptable
  - Lab preparation **(~1 page + additional material (test cases, etc.))**
    - Description of the laboratory setup (materials, instruments, software)
  - Lab session **(~3 pages)**
    - Presentation and discussion of the results
    - Source code and circuit diagrams should only be shown in excerpts if they are necessary for explanation.
    - Source code and large drawings can be put into the appendix
  - Conclusion **(~1/2 page)**
    - Summary of key findings at the end of the report
  - Literature
    - List of literature and sources
  - Appendix
    - Source codes and other documentation that do not fit into the main body of the text (e.g. handwritten notes of measurement results).
    - Please limit the printing of source code to your own parts or to relevant modifications of original source code.
    - Use text boxes to display source code and use a monospace font, such as Courier, in a maximum size of 9 points.

## References and Plagiarism

- Report structure and any reference to external sources, such as texts, images or source code, etc must be correctly referenced.
- Unmarked quotations (including image quotations) will result in your report being classified as plagiarism.

**Plagiarism will result in failure of the laboratory (for the entire semester and for the entire team!)**

## Scoring system

- There is a scoring system for assessing laboratory tests.
- In total, 165 points can be reached

| <i>Exercise</i> | <i>Max Points</i> | <i>Comment</i>                                                 |
|-----------------|-------------------|----------------------------------------------------------------|
| 1               | 18                | <i>Protocol only. No preparation required</i>                  |
| 2               | 47                | -                                                              |
| 3               | 58                | -                                                              |
| 4               | 42                | <i>No report required. Title Score board must be submitted</i> |
|                 | 165               | <i>/108 needed</i>                                             |

- To pass the PVL you need at least
  - 108 points (~65%) in total
  - 30% of the max points for each exercise
- If you have collected enough points to pass the PVL at the last appointment, you still have to attend to the lab sessions
- Students who arrive late will be penalized 1 point for every 2 minutes of delay. This applies to delays of more than 3 minutes, with the first 3 minutes not counting. So, for example, if you arrive 14 minutes late, you lose 5 points.

**Example – Title page Exercise 1**

## Digital Circuits

### Exercise 1- Measurements

|                              |                        |
|------------------------------|------------------------|
| <b>Group/Team No.:</b> _____ | <b>Date:</b> _____     |
| <b>Name :</b> _____          | <b>Name :</b> _____    |
| <b>Prenome :</b> _____       | <b>Prenome :</b> _____ |
| <b>IM No.:</b> _____         | <b>IM No.:</b> _____   |

| 1st lab exercise                 | max score | score reached |
|----------------------------------|-----------|---------------|
| Security training                | mandatory | 0             |
| Preparation                      | 0         | 0             |
| Task 1 (static behavior)         | 2         |               |
| Task 2.1 (generator adjustment)  | 2         |               |
| Task 2.2 (rise/fall time, delay) | 3         |               |
| Task 2.3 (delay)                 | 1         |               |
| Task 3.1 (glitch)                | 3         |               |
| Task 4.1 (oscillator)            | 2         |               |
| <b>Sum</b>                       | <b>13</b> |               |
| <b>Protocol</b>                  | <b>5</b>  |               |
| <b>Penalty</b>                   |           |               |
| <b>Total points</b>              | <b>18</b> |               |

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**Example Score board – Exercise 2/3/4**

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